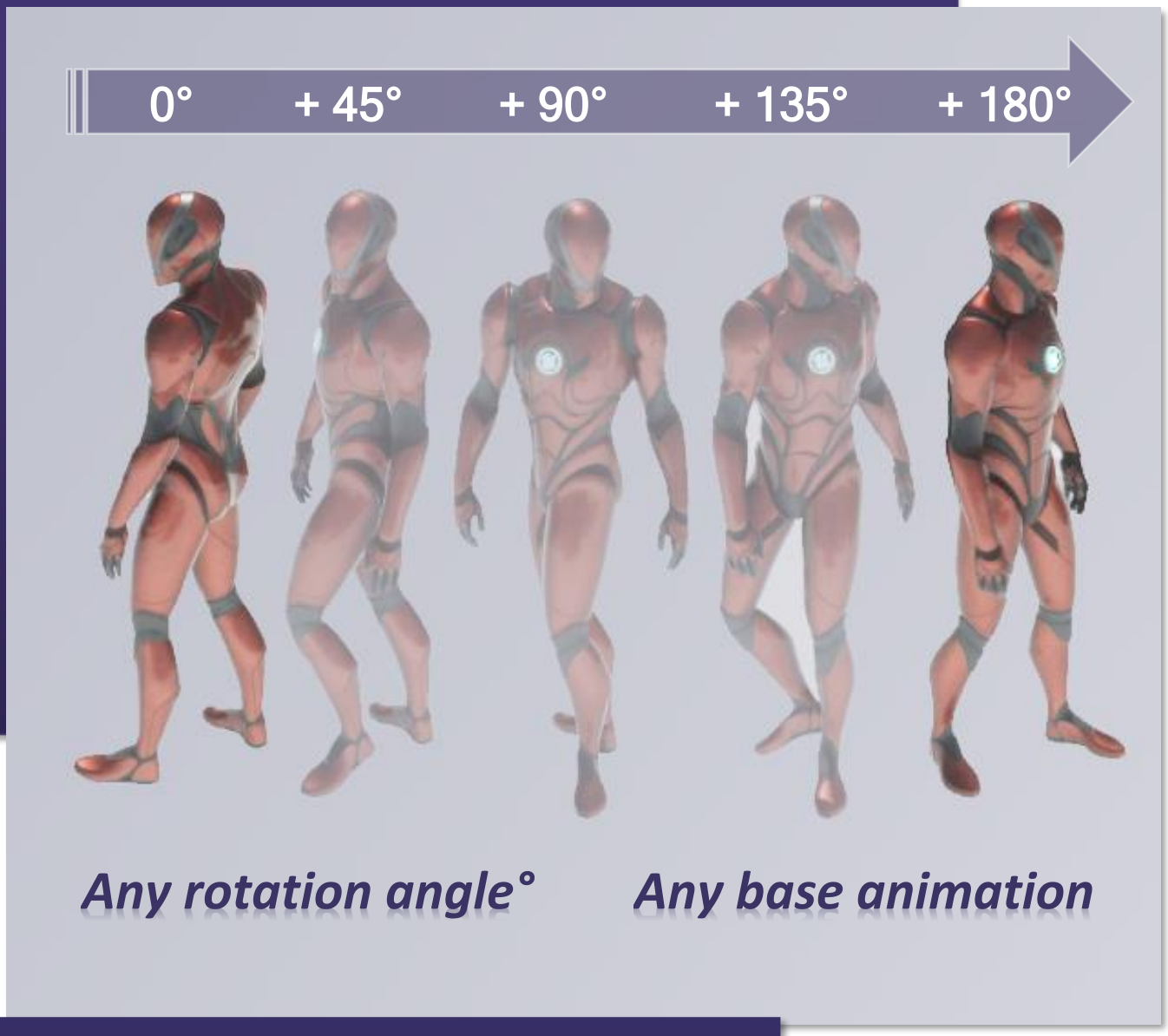

Procedural Turn in Place System

Far South Creations

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INTRODUCTION

Thank you for your purchase. The Procedural Turn in Place system consists of a Control Rig node + an Actor component that will automatically generate in the fly animations for turning in place for any angle.

It solves a difficult problem present in other Turn in Place systems which 1) either look great (using root motion) but are restricted to specific angles, 2) allow any angle but with noticeable feet sliding or 3) require very specific 360° animations and complex blueprint setups. This system, on the other hand, will take any pre existing idle pose and calculate how many steps are required to fully rotate to the required angle (which can be anything from -180 to 180) and then predict the best foot position to for these steps. It will also shift the characters Center of Mass to ensure it looks realistic and to prevent feet overextension.

Issue	Other systems			Procedural Turn In Place
	Root Motion	Manual Rotation	Pose-Evaluation	
Allow rotation for any angle	✗ Yes	✓ Yes	✓ No	✓ Yes
Have unsightly feet sliding	✓ No	✗ Yes	✓ No	✓ No
Require at least 2/4 custom animations	✗	✗ Yes	✓ No	✓ No
Require 360° animation	✓ No	✓ No	✗ Yes	✓ No

Below you'll find a quick start guide as well as detailed explanations on customizing the system. Please also note that you can head over to my youtube site for video tutorials.

If you require additional support, please contact me at support@farsouthcreations.com and I'll do my best to help you out. Please be patient as I'm a single developer and might not always be able to reply on time.

If you're happy with this product, please consider reviewing it in the UE5 marketplace. Reviews really help with projects visibility and allow me to continue updating assets, solving bugs, and providing dedicated support. If you're not happy with the product, please let me know and I'll do my best to help and, if all else fails, reach out to Epic for a refund.

Thank you

UPDATE HISTORY

- **v1.0** 11-Jan-2023
  - Initial release for UE 5.1
- **v1.1** 16-Jan-2023
  - Added support for UE 5.0 + Project cleanup
- **v1.2** 27-Jan-2023
  - Added network replication
  - Added footstep functionality examples
  - Simplified integration
  - Fully redesigned the ControlRig and TurnInPlaceComponent code to bypass a UE5 bug that prevents “SetVariableFromString” from working in some packaged projects

QUICKSTART GUIDE

Using the default character

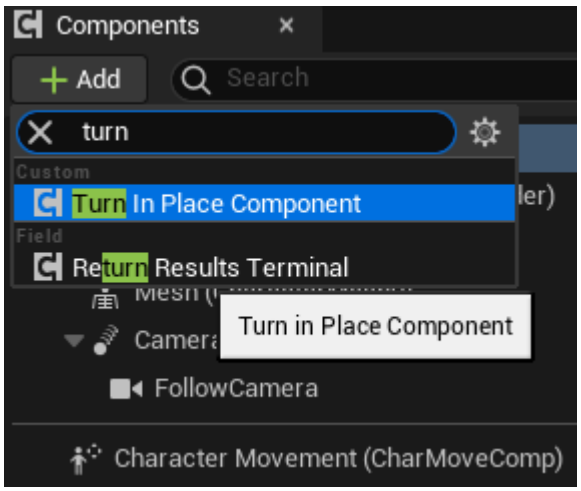
- In the “CharacterBlueprints” folder you’ll find several ready-to-play Characters both using the UE5 (recommended) and UE4 Mannequins. Open the “TurnInPlace_Showcase” map to test these characters and see if you can start from there.
- The “BP_UE5_ThirdPersonCharacter” uses the default ThirdPersonTemplate character with the additional TurnInPlace integration.
- By default, if you hold Right Mouse your character will strafe and automatically rotate towards the camera direction while running.
- If you rotate the camera while standing still, the character will TurnInPlace procedurally whenever the Angle between the Actor Rotation and Control Rotation (i.e. camera) exceeds 45°.

Implementing the Procedural TurnInPlace system in other Characters

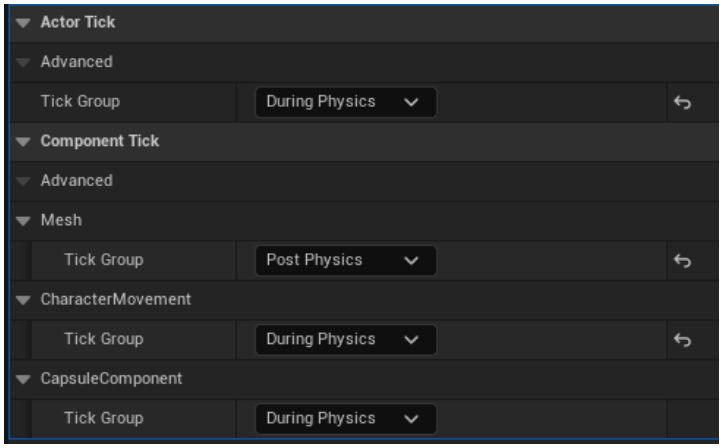
Tip: you can see step-by-step video tutorials here.

The easiest way to implement this in other character-based projects is to do the following:

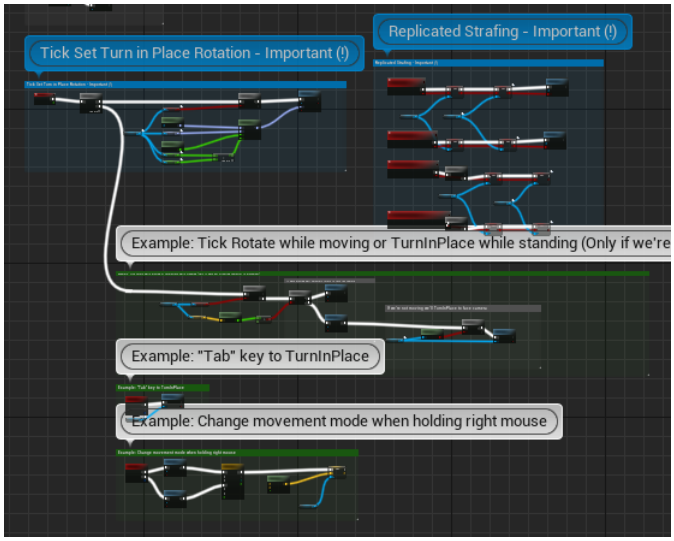
1) Open your CharacterBlueprint and click “Add” > “TurnInPlaceComponent”. If you click in the component you can customize movement parameters. Refer to the Turn in Place Component in Depth section for more information.



2) In the “Class defaults” of your Character Blueprint search for “Tick group” and set the Mesh tick group to “Post Physics” and the others to “During Physics”



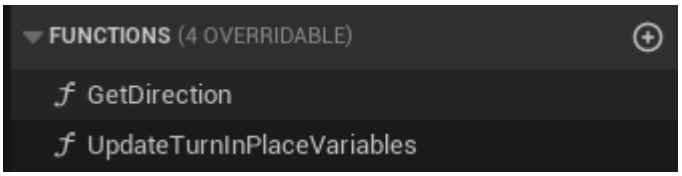
3) Open the BP_UE5_ThirdPersonCharacter found in the “CharacterBlueprints” folder. Copy the code commented in Blue (mandatory) and Green (optional) to your character blueprint:



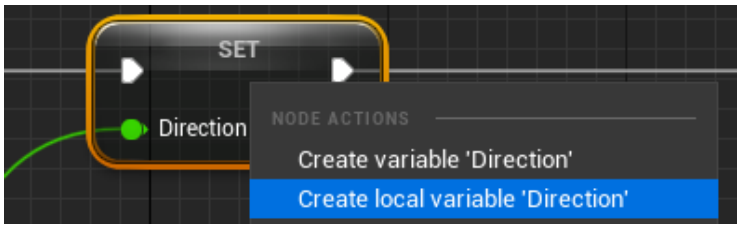
4) Inside the “Rotate Character Smoothly” event make sure to right click in the “Target Actor Rotation Replicated” > “Create Variable”. In the details pane set it to “Replicated”:

5) Open your character’s Anim Blueprint and also the “UE5_ABP_MannyExample” asset (or the “UE4_ABP_Example” if you’re using the UE4 Mannequin) found in the AnimBlueprints. Then we have to start copying some functionality over to your AnimBlueprint:

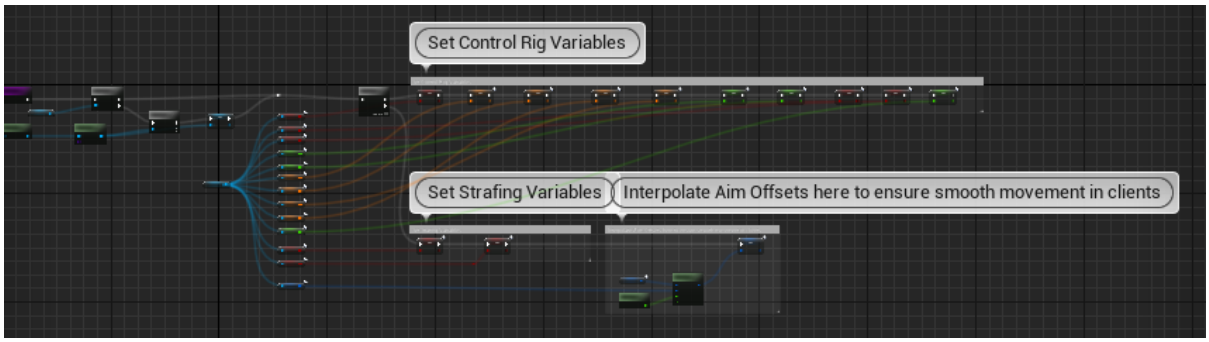
   a. Copy the “GetDirection” and “UpdateTurnInPlaceVariables” functions on the left bar from the Example Anim Blueprint over to your own Anim Blueprint:

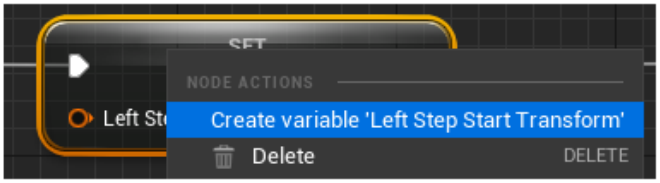


b. Open the “GetDirection” function, right click on “Direction” > “Create variable”.

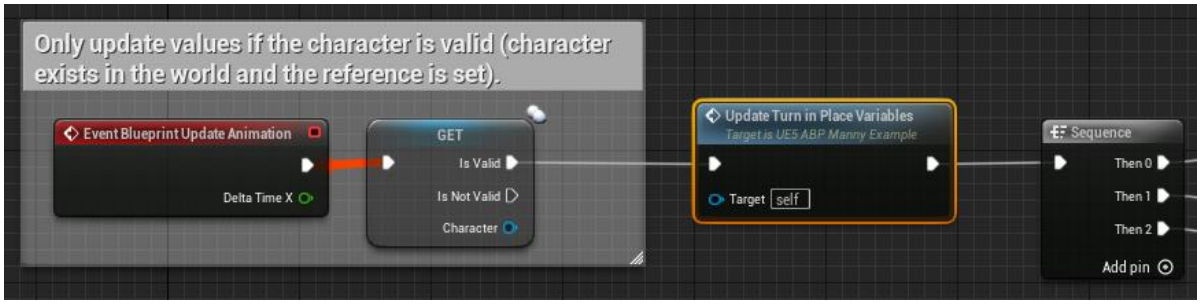


c. Open the “UpdateTurnInPlaceVariables” and right click on each one of the **14** variables > “Create Variable” and set them as “Replicated”:

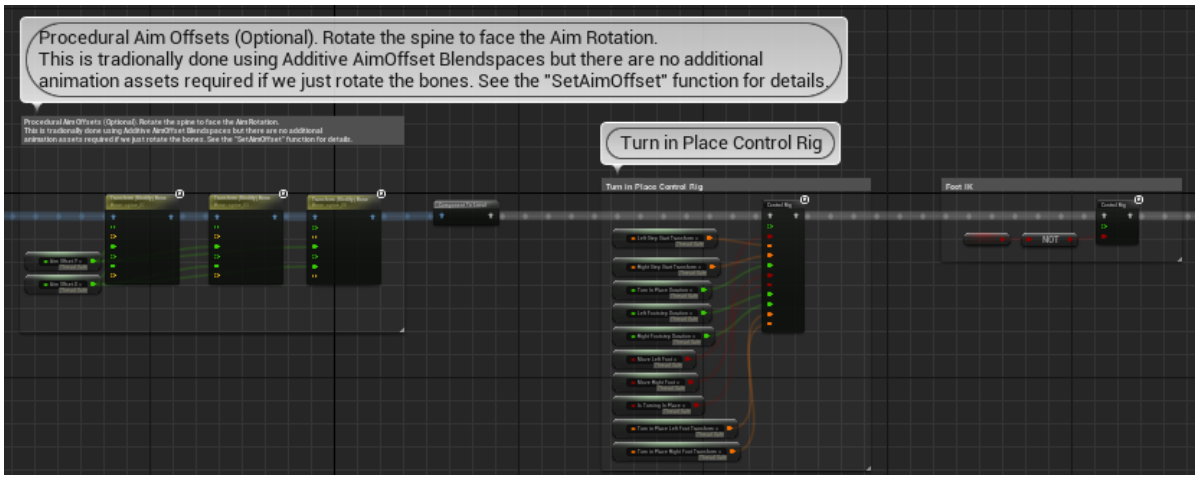




d. In your AnimBlueprint’s “Event Blueprint Update Animation” add a call to the “UpdateTurninPlaceVariables” function:



e. Go to the AnimGraph and copy the Aim Offsets, Turn in Place Control Rig, and the Feet IK (optional but recommended if you’re using the default IK) to your AnimGraph:

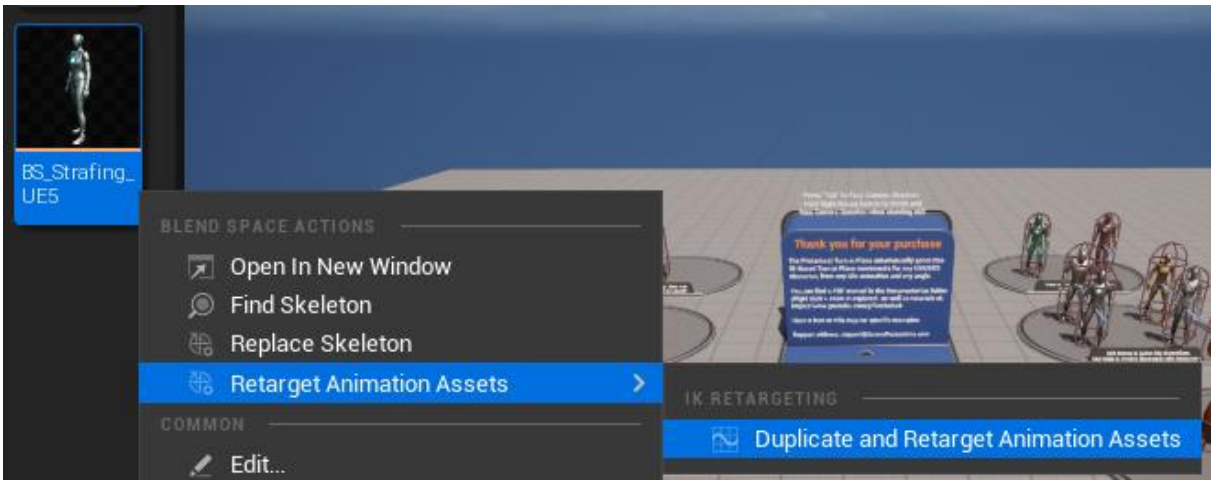


6) Enjoy

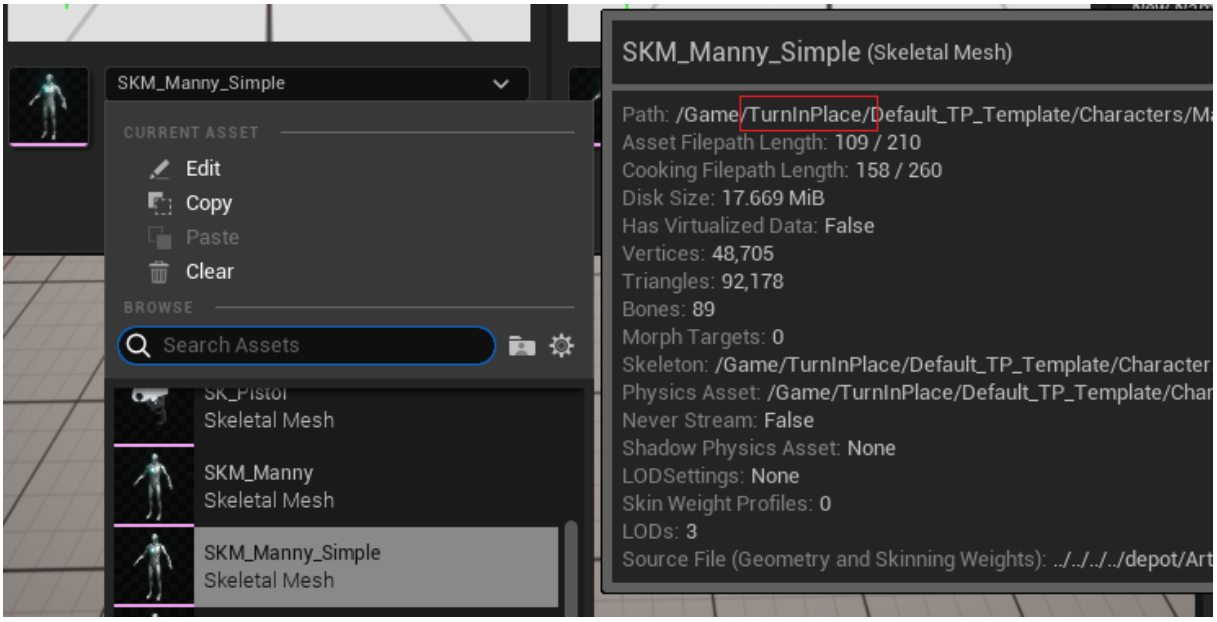
Adding Strafing Animations (Optional)

The system includes, by default, 8 Strafing animations. If you want to use them in your own character you'll have to retarget them first. It's a simple process:

1) Right click on the "BS_Strafing_UE5" Blendspace > "Retarget Animation Assets"

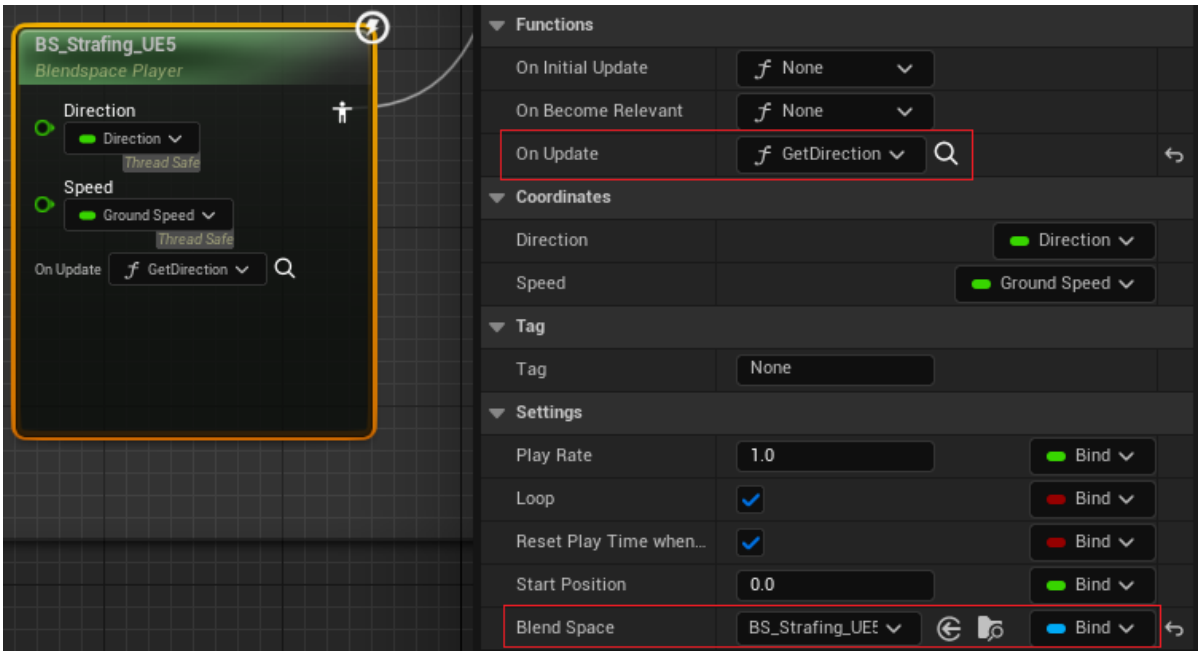


2) Select "RTG_Mannequin" in IK Retargeter, and make sure to select the SKM_MannySimple Skeletal Mesh that is in the TurnInPlace directory in "Source" and your own Manny Skeletal mesh in "Target":



3) After retargeting, go to your own AnimBlueprint and, in your Walk/Run state, add the blendspace that you just retargeted. Make sure to connect the "Direction" option to the "Direction" variable, and do the same with the "Speed" option. Finally, make sure to add the

“GetDirection” function to the “On Update” Section.

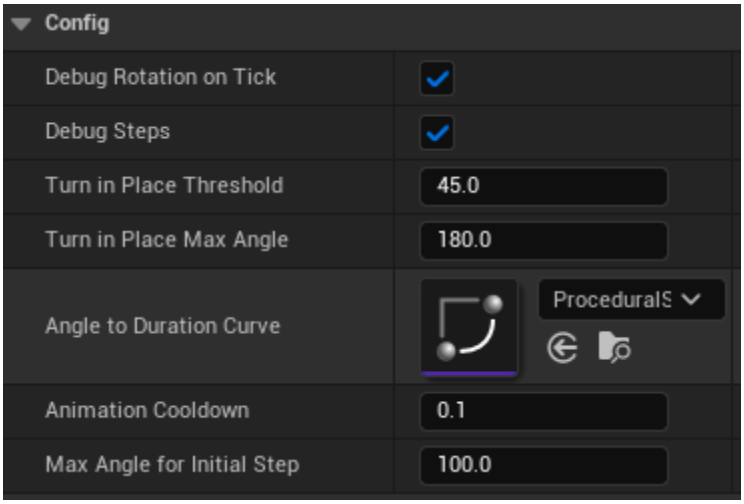


4) That’s it – If you now hold the right mouse button, you’ll start strafing.

TURN IN PLACE COMPONENT IN DEPTH

Main Configuration

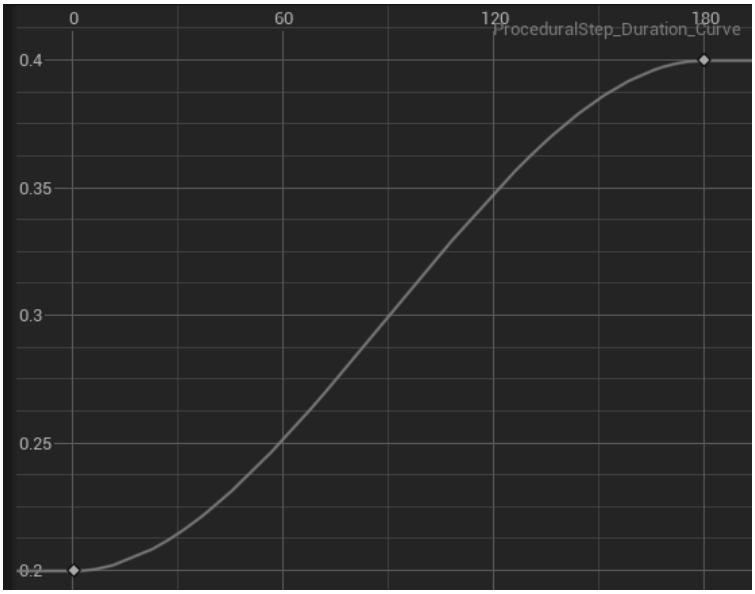
If you click in the Component you can customize the movement parameters.



1) The Debug Rotation on Tick will show two arrows in front of the player and a number showing the angle between them. The green arrow represents the Control Rotation (which is the camera orientation) and the blue arrow represents the current Actor Rotation (where the character is really facing).
2) The Debug Steps will show the predicted step locations whenever TurnInPlace is triggered and will also print a string detailing the movement.
3) The Turn in Place Threshold is the minimum angle in which the TurnInPlace process will trigger (Lower values are better handled by Aim Offsets).
4) The Turn in Place Max Angle can be used to limit how much a single rotation can be. By default characters can rotate at any angle (-180 left or +180 right), but you can limit this if needed.
5) The Angle to Duration Curve defines how long, in seconds, it will take for each step according to the angle traversed. There are two curves included by default. In the “Showcase” map you can see that there is a Giant UE5 Mannequin that uses the slow curve (and takes roughly twice to turn). You can create any number of curves as required for your characters.
6) Animation Cooldown specifies how long to wait between each animation. The number should be at least for 0.05 seconds.
7) The Max Angle for Initial Step minimizes collisions by limiting how far the first step can go. 100 seems to work for almost all animations, but if you find any collision if your animations you can try lowering the value.

Changing Step Duration

You can define exactly how long steps take by changing the “Angle to Duration Curve” in the TurnInPlace Component. The default curve looks like this:

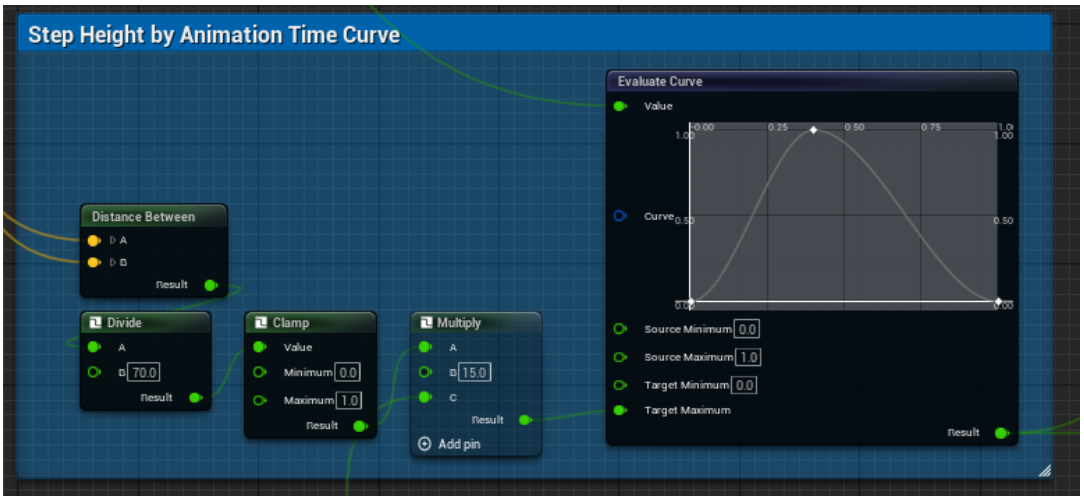


Which means that steps traversing 180 angles (full half rotation) will take 0.4 seconds, while smaller angles take between 0.2 and 0.4.

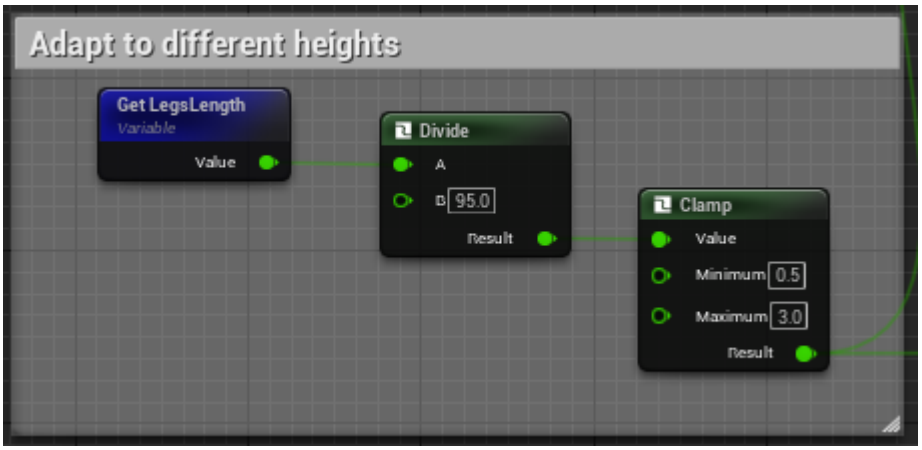
Changing Step Height

The “CR_TurnInPlace_UE5” Control Rig at the System>ControlRig folder contains the entire logic defining how high steps go and the timing to get to their targets:

Inside the ControlRig, click in the “Step” function to edit these curves as required:



By default, the feet will reach its highest point at 39% of the Step Duration. The height will be defined by the distance traversed by the feet, in which 70 unreal units (70 cm) will mean lifting the feet 15 cm from the ground. You can change any of these figures to change the step height.
In addition, note that the height will also depend on the character's legs size, as defined in the "Adapt to different heights" code block:



The default UE5 mannequins have legs that measure roughly 95 cm, so we'll divide the Legs Length of your mesh (calculated as soon as the Rig is initialized) by 95 and multiply that for the Height Calculated in the previous step. This means that if you have a mannequin that is twice the size of the standard mannequin, feet will be lifted by twice the amount of a non-scaled mannequin.

The basic formula looks like:

$$\frac{distance}{70} * 15 * \frac{Legs\ Length}{95} * TimeCurve$$

Green clamped to (0 to 1)

Blue clamped to (0.5 to 3)

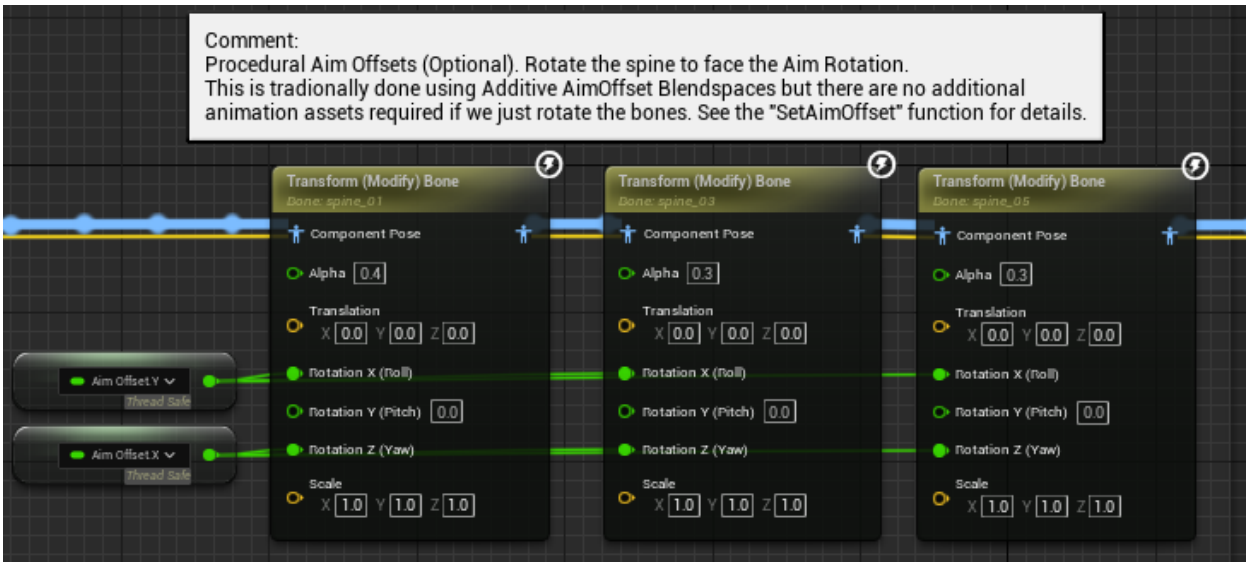
The previous figures are just arbitrary numbers that looked great for me while testing several animations, but feel free to edit them as desired if you think it will adapt better to your game.

Please note that UE 5.0 version has minor differences in the ControlRig made to correct engine bugs no longer present in the UE 5.1 version. They work very similarly but the code is somewhat different.

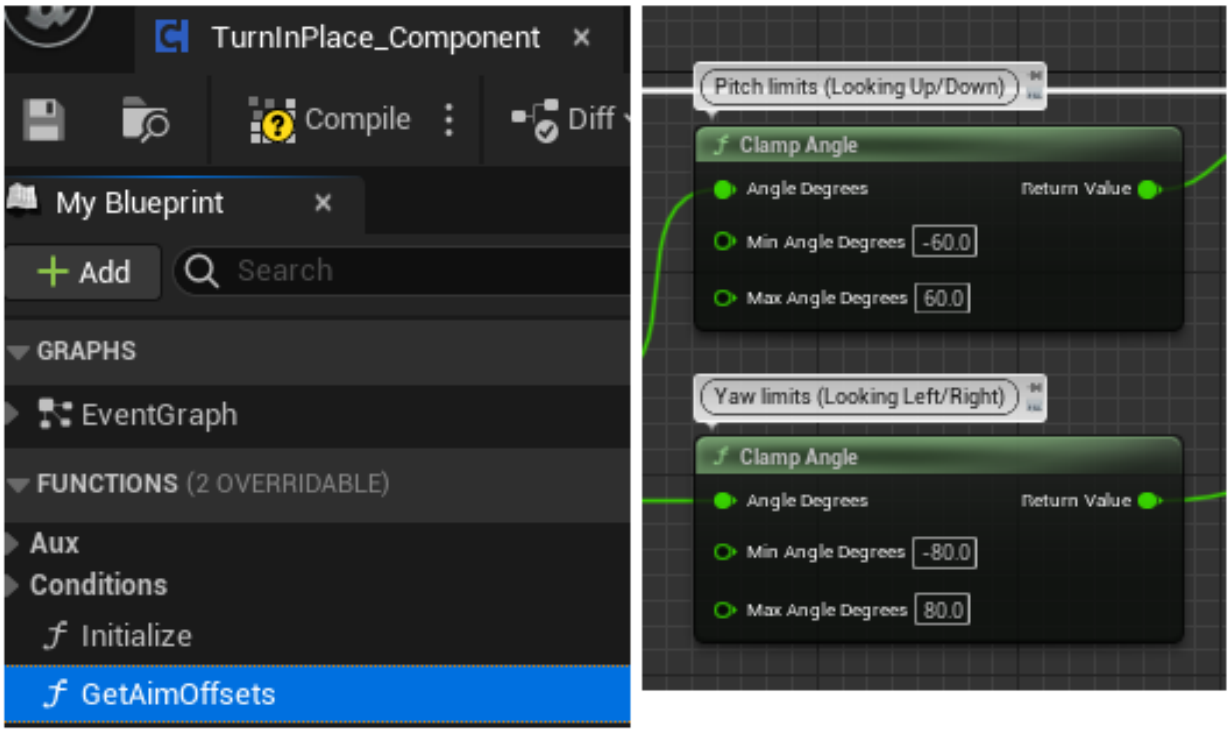
About Aim Offsets

The standard Unreal Engine practice is to use "Aim Offset Blendspaces" to rotate the upper part of character's Torso while keeping feet planted in the ground, to make the character face different rotations as shown in unreal's documentation. This requires a lot of animations (the example documentation shows 15). To provide a simpler solution, I added some nodes to the Anim Blueprints that just rotate the Spine bones to imitate this with no animation assets required. It

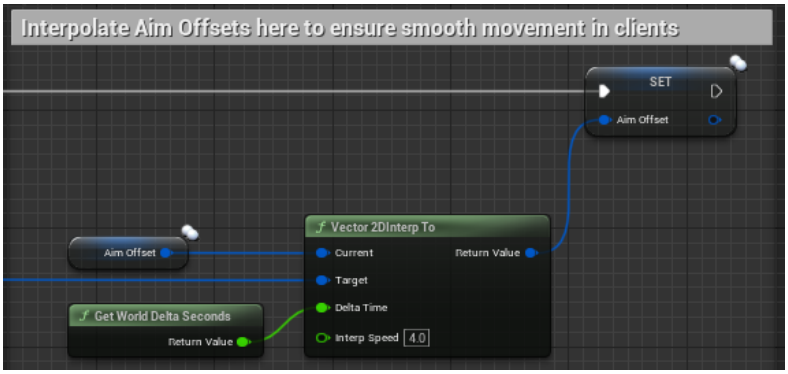
works great for most cases and doesn't require any animation assets:



You are free to use this or your own AimOffset solution. In case you have your own, just delete these AimOffset nodes. If you want to change the rotation limits, you can do so by opening the "**GetAimOffsets**" function inside the **TurnInPlaceComponent** and setting it here:

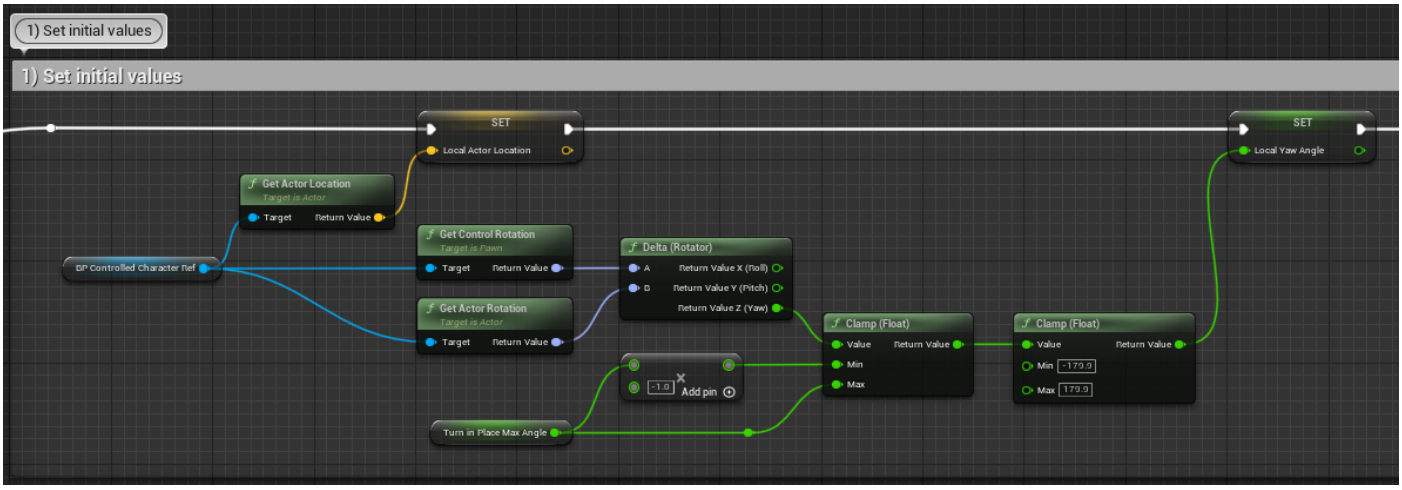


If you want to change the speed in which the character Aims, you can do so inside the "UpdateTurnInPlaceVariables" here:



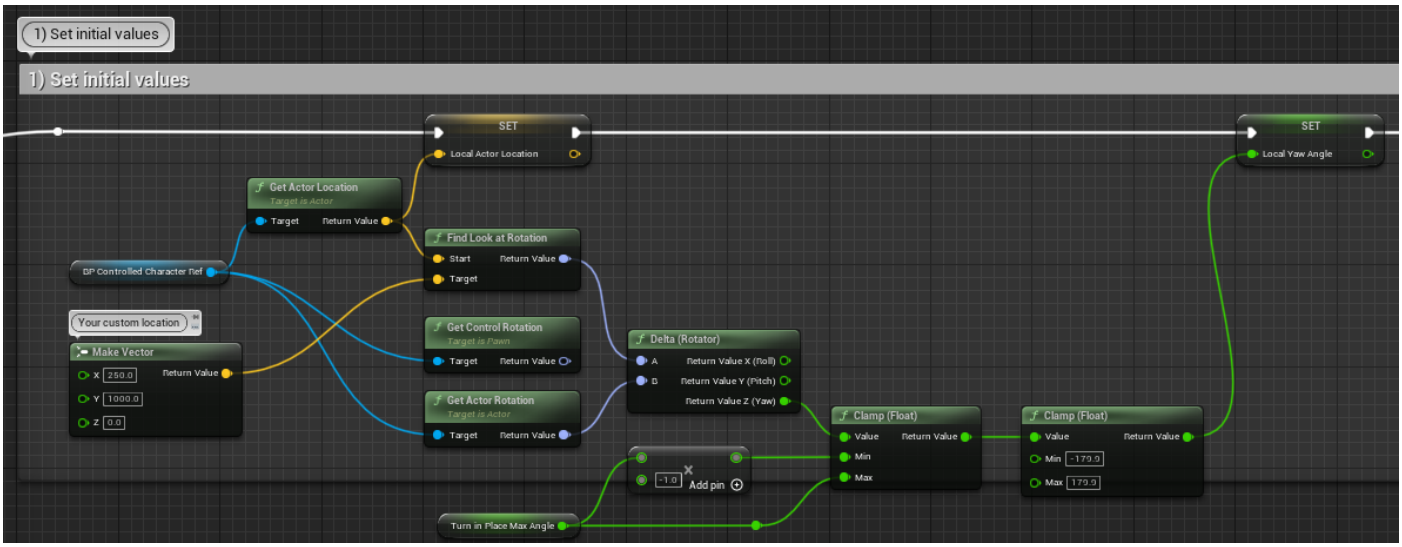
Changing the target rotation

By default, the Turn In Place system target rotation will be the “Control Rotation”, which is usually the Camera Rotation. This is defined in the “CalculateTurnInPlaceSteps” function, here:



In the above screenshot we are calculating the delta (difference) between the Control Rotation and the Actor Rotation. This defines the “Yaw Angle” which is basically how much the character will rotate. If you want to change this and make the character face a different rotation, make sure to replace the “Get Control Rotation” node with your own rotation instead.

For example, if you wanted to look at a specific world position (let’s say we want the character to face World Location X=250 Y=1000), you might change it as follows:

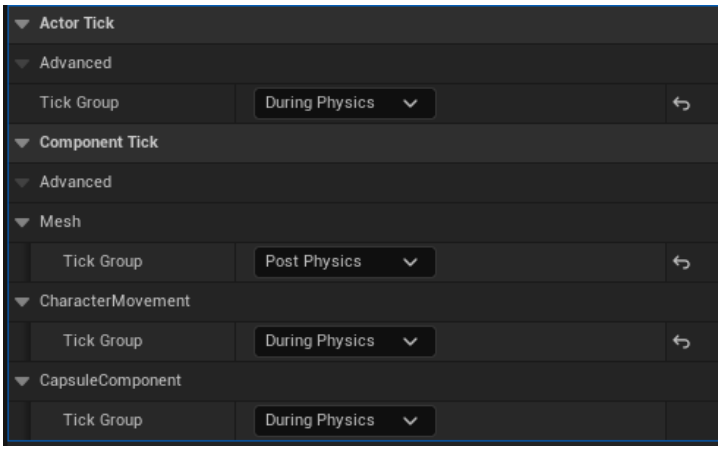


You can use this technique to face arbitrary rotations and for other movement schemes such as top-down games (In which case you’ll want to face the mouse-to-world location)

TROUBLESHOOTING

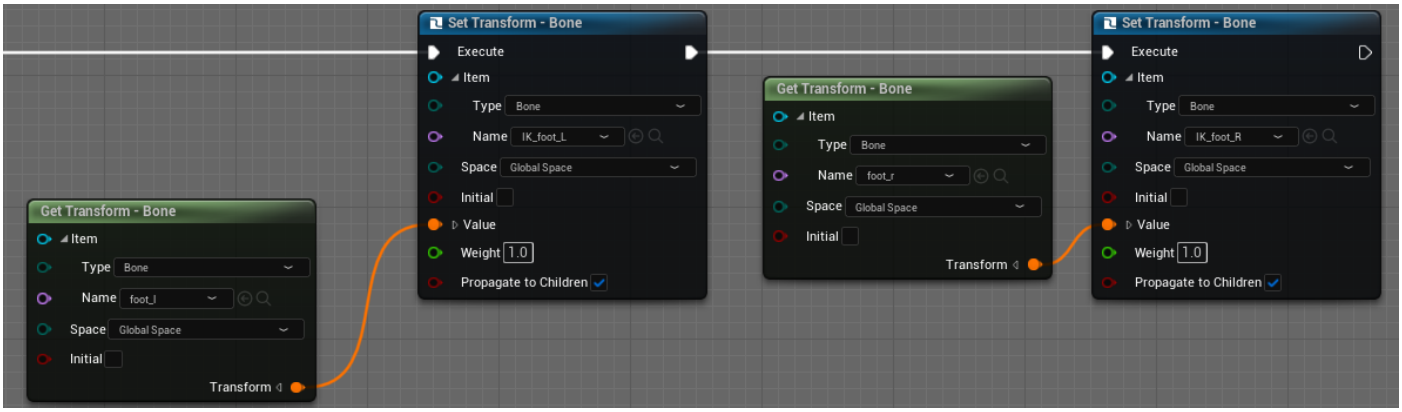
The is some small jittering sometimes when starting the animation

Make sure the Mesh Tick Group is set to “Post Physics” and the others to “During Physics”:



I am using the default UE5 IK and the feet don’t move when changing rotation

This is an issue with UE5’s default IK which doesn’t set the ik foot bones to their position properly. The easiest fix is to simply use the IK included by default in the Turn in Place Manny ABP. If you want to solve it yourself, simply add these nodes to your IK ControlRig at the beginning:



Turn In Place motions don’t show in all Clients or Server

Make sure to set the required variables to “Replicated” following the Quickstart Guide.

I set this up in a First Person Character but it’s rotating automatically in a ‘glitchy way’

Make sure that “Use Control Rotation Yaw” is turned off.